Environmental Science & Technology

pubs.acs.org/est

Article

Open Access

This article is licensed under CC-BY-NC-ND 4.0

A High-Precision Geo-Spatial-Techno-Economic Framework for CO_2 Source-Sink Pairing: Land-Sea Synergy and Cross-Regional Coalitions in China

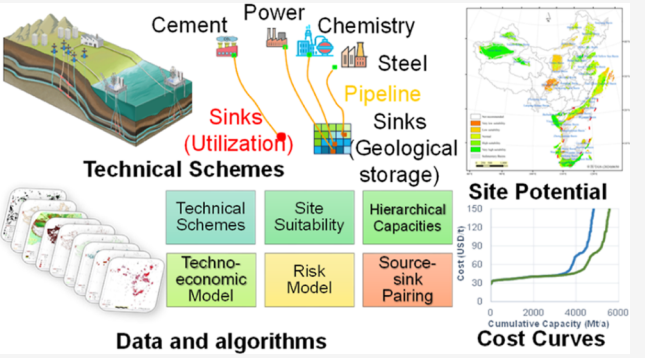
Ning Wei,* Keyao Lin, Shengnan Liu, Dalin Jiang, Wendong Wang, Zunsheng Jiao, Robert T. Dahowski, and Xiaochun Li

Cite This: *Environ. Sci. Technol.* 2026, 60, 19170−19184

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: China suffers from a severe geospatial mismatch between industrial and energy CO_2 emission sources and geological storage sinks, while existing carbon storage assessments lack national-scale high-resolution and land-sea synergy analysis. This study develops a high-resolution (1 km × 1 km) techno-economic framework integrating hierarchical storage capacity, site suitability, and full-chain carbon capture and storage (CCS) source-sink matching to clarify China's geological carbon storage characteristics. The results show nearly balanced onshore-offshore theoretical (2.5 trillion tons) and P50 effective (1.5 trillion tons) storage capacities, with distinct cost gaps: onshore storage costs of 3−10 USD/t and offshore costs of 10−60 USD/t. Benefiting from regional capacity and transportation economic heterogeneity, 60−75% of China's coal-based captured CO_2 achieves a levelized cost below 90 USD/t. This fine-scale framework maps national CCS distribution heterogeneities across multiple scales. We further propose cross-regional CCS coalitions via infrastructure sharing, land-sea storage integration, and cross-region carbon corridors to mitigate imbalance and could manage about 80% of coal-sector emissions, providing a scalable solution for Global South countries with unbalanced carbon development.



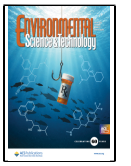
KEYWORDS: *zone-scale framework, source-sink pairing, hierarchical capacity, techno-economic model, levelized cost, regional heterogeneity*

1. INTRODUCTION

Carbon capture, utilization, and geological storage (CCUS) emerges as a linchpin for the decarbonization odyssey of fossil-intensive industries and energy sectors, including thermal power, cement, chemical, and steel production, in their pursuit of carbon neutrality.[1−6] At the heart of successful CCUS scaling is the meticulous evaluation of storage sites, a prerequisite for garnering stakeholder confidence, particularly for CO_2 deep saline aquifer storage, which accounts for more than 90% of the total contribution. However, the intricate geological heterogeneity of these formations, coupled with diverse trapping mechanisms operating across disparate temporal and spatial scales, exacerbates the challenge.[7−9] Unlike oil and gas reservoirs, which benefit from decades of comprehensive site characterizations and operational data, deep saline aquifers suffer from a dearth of comparable data sets. This severely limits empirical insights and hinders stakeholders' efforts to accurately quantify storage capacities, assess costs, bridge source-sink mismatches, and evaluate associated costs.[9,10]

Although significant advancements in capacity evaluation have been made, inconsistencies and uncertainties persist, stemming from divergent technical assumptions, variable data quality, and fragmented assessments of basins. In China, the large-scale deployment of onshore carbon capture and storage (CCS) faces unique hurdles, primarily due to the pronounced spatial disparity between CO_2 emission sources and viable geological storage reservoirs.[11−14] This geographical mismatch imposes very high cost of transport and storage, e.g., >60 USD/t CO_2, and creates substantial regional cost gaps between northern/inland storage hubs and southeastern/eastern emission-intensive centers in the Southeastern Coastal and South China regions. Relying solely on onshore storage in the distant Three Norths (Northwest, North, and Northeast) regions renders mismatched projects economically unfeasible.[11,12,15] Although scholars have conducted disjointed evaluations of

Received: October 14, 2025
Revised: June 17, 2026
Accepted: June 18, 2026
Published: June 30, 2026



Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

ACS Publications

© 2026 The Authors. Published by American Chemical Society

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

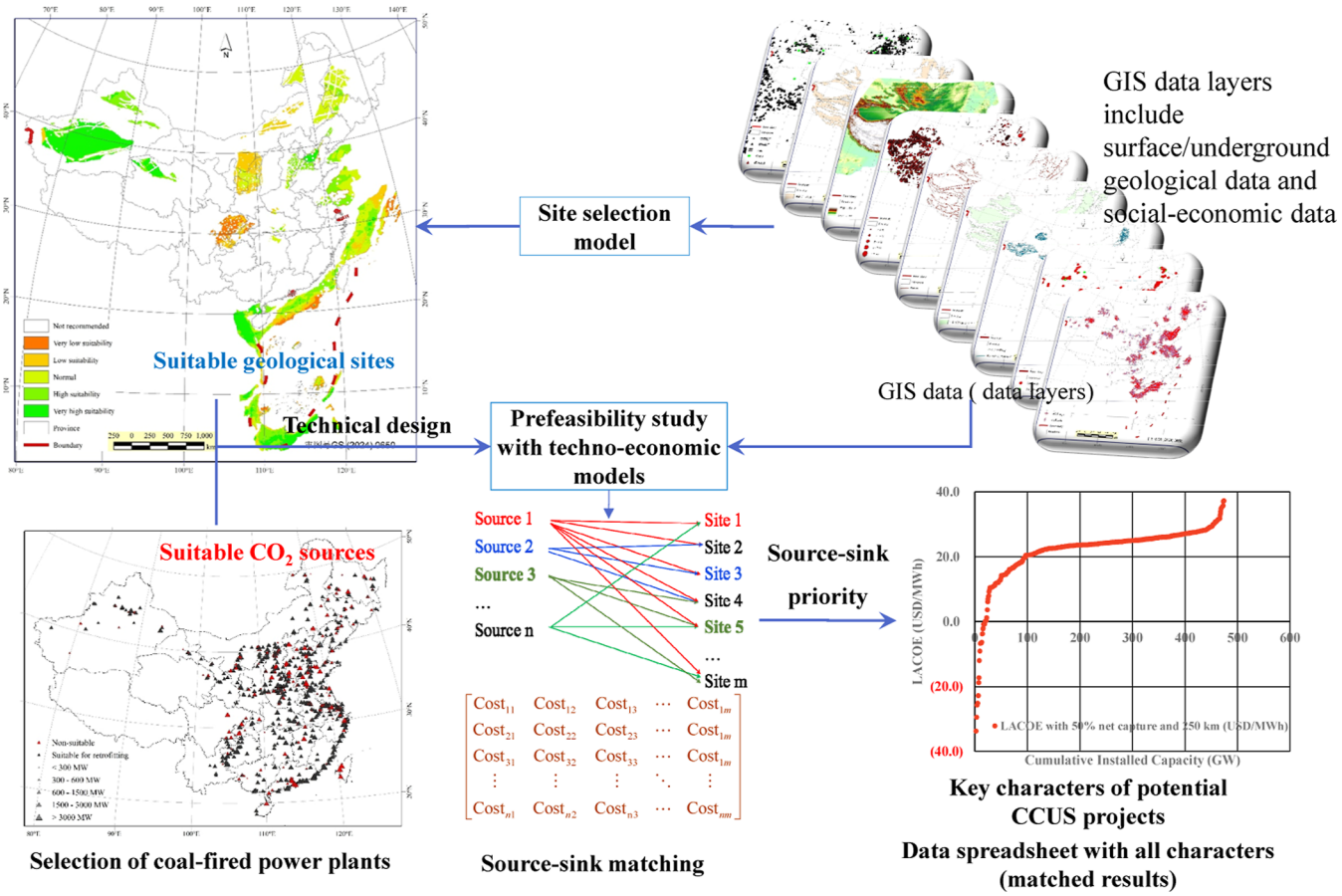


Figure 1. Schematic figure of zone-level framework source-sink pairing for CO_2 capture and storage.

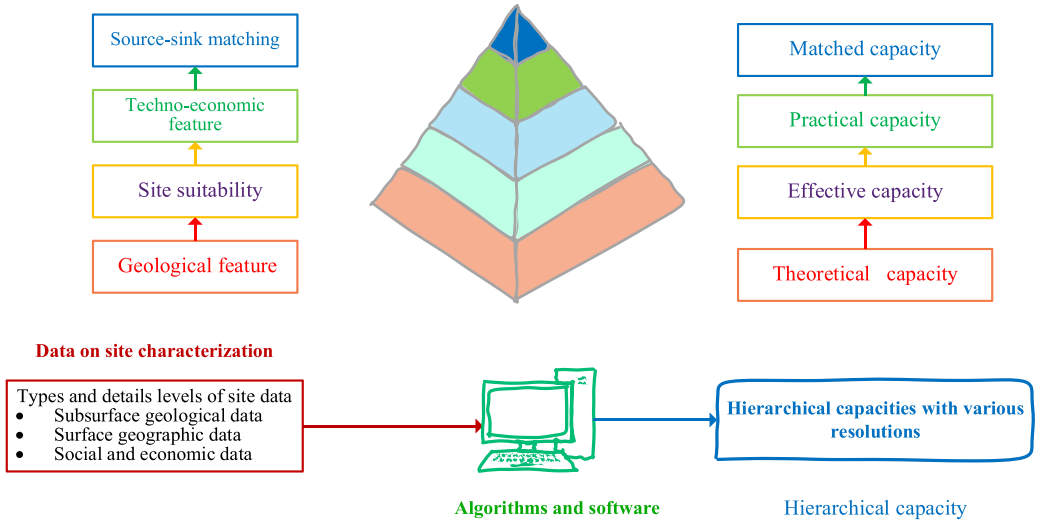


Figure 2. Tiered geological-economic assessment with various algorithms and data.

capacity and site suitability in selected onshore (e.g., Junggar, Songliao) and offshore basins (e.g., Pearl River Mouth, Bohai Basin, Beibuwan), inconsistencies in data quality (geological, geographical, and socio-economic) and methodologies impede their integration and comparison.[16−21]

Various qualitative and quantitative algorithms and methods are utilized for capacity assessment, site suitability evaluation and selection, techno-economic analysis, source-sink matching, and quantification of mitigation potential. Conventional source−sink matching primarily aligns the theoretical storage capacity with CO_2 emission sources, while techno-economic evaluation mostly relies on empirical or oversimplified models. A critical limitation in existing studies lies in the poor integration of geological storage potential, techno-economic performance, source−sink configuration, and macro policy constraints.[22−24] Furthermore, the lack of high-resolution, standardized multi-dimensional assessment frameworks that couple capacity, site suitability, cost, and policy—together with ambiguous data

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

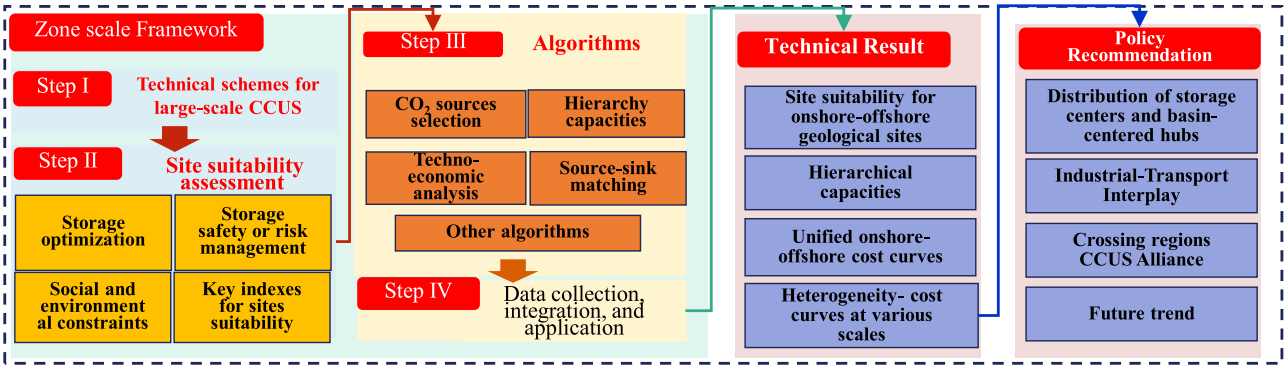


Figure 3. Schematic figure of the zoning evaluation framework and analysis.

quality specifications—prevents robust national-scale holistic evaluation.[2,25,26] This gap impedes coordinated CCS planning across sectors, regions, and land−offshore boundaries.

Site assessment for CO_2 geological storage is commonly conducted at basin, sub-basin/zone, and individual site scales, depending on data type, accuracy, and methodological algorithms.[17,27−30] Storage capacity evaluation generally covers four hierarchical dimensions: theoretical, effective, practical, and matched capacity, with assessment approaches evolving from theoretical extrapolation to refined actuarial estimation for improved accuracy.[27,28] The site suitability assessment and theoretical capacity evaluation of offshore CO_2 geological storage in China primarily focus on individual basins or theoretical capacity assessments at the basin scale,[29,30] without considering the synthetic effects of capacity, suitability, cost, and policy. These high-resolution assessments are data-driven by a high-quality database (data type and resolution). The tiered geological-economic assessment with various algorithms and data are shown in Figure 2. Notably, zone- and sub-basin-scale evaluations of site selection and tiered storage capacity are still limited, especially for China's offshore regions.

These shortcomings are further exacerbated by China's incomplete onshore−offshore geological data sets, basin-isolated and fragmented site assessment methodologies, generalized one-size-fits-all techno-economic models, assumption-dependent source−sink matching schemes, and emission inventories confined to individual sectors or limited regions. Collectively, these prevailing limitations constrain a systematic understanding of national CCS development potential and industrial deployment pathways, hinder integrated evaluation of storage capacity, economic feasibility, source−sink optimization and policy adaptability across multiscale administrative and geographic units, and may ultimately mislead stakeholder decision-making and strategic CCS layout.

This study establishes a nationwide 1 km × 1 km high-resolution evaluation framework that systematically integrates zone-level onshore−offshore storage suitability, hierarchical capacity quantification, techno-economic models of CCS projects, enterprise-level carbon emission accounting across multiple industries, and source−sink matching across the full CCS chain (capture, pipeline transportation, and geological storage) based on pipeline cost surfaces, and carbon reduction contribution forecasting, enabling comprehensive analysis at the kilometer scale and beyond. Figure 1 illustrates the schematic of the zone-level framework. A database of CO_2 sources, geological features, geographic characteristics, and related socioeconomic data in China and adjacent areas was established utilizing a wider range of high-precision data. Then, algorithms of CO_2 source screening, site suitability assessment, hierarchical storage capacity, techno-economic assessment, source-sink pairing, and pipeline network generation algorithms were selected and integrated into a software named "Integrated Techno-Economic Assessment Model of Carbon Capture, Geological Utilization, and Storage with hub, ITEAM-CCS hub".[12,31,32] The framework addresses inherent limitations of conventional basin-scale and sector-specific assessments by supporting national-scale cross-regional and cross-sectoral comparative analysis. It further quantifies spatial heterogeneities across basins, geographical regions, industrial sectors, and administrative regions, and delivers practical pathways for CCS collaboration, integrated land-sea management, and cross-regional carbon corridor planning to facilitate evidence-based policy formulation and large-scale CCS deployment. Additional details regarding the ITEAM-CCS model can be found in the papers by Wei, et al.[11,12] It unlocks the intricacies of China's CCS landscape, providing a comprehensive understanding across sectors, regions, basins, and land-sea boundaries, heralding a new era in CCS planning and implementation.

2. METHOD AND DATA

High-precision evaluations of geological structures, techno-economic factors, and carbon-neutrality targets are crucial steps toward achieving large-scale deployment of carbon geological storage and achieving neutrality goals. However, there is currently a scarcity of publicly available assessment reports on site selection and storage capacity for CO_2 geological storage at the sub-basin and zone scales. This is particularly true for zone-scale assessments and studies on onshore/offshore storage capacities in China. Through data compilation and algorithm development, we propose a high-resolution framework for site selection, hierarchical capacities, economic analysis, and source-sink pairing. This involves a multiple-stage assessment integrating technical scheme definition, suitability, algorithms, data compilation, and data-driven analysis at ∼1 × 1 km^2 grid resolution (Shown in Figure 3). This zone-scale methodology complements site-scale tools (e.g., Petrel, TOUGH2, MRST, et al.) by geospatially integrating storage capacity, site suitability, and economic viability via stratigraphic analysis, cost-surface modeling, and multiobjective optimization. Unlike traditional meter-to-hundred-meter-resolution single-reservoir models, our kilometer-gridded system enables national-scale CCS infrastructure planning by identifying strategic storage hubs and optimizing cross-basin source-sink pairings. This shift from isolated project analysis to systematic evaluation addresses critical scale-resolution mismatches in carbon management policies, providing both technical outputs (site suitability,

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

capacity hierarchies, cost curves) and policy guidance for regional potential distribution and transitions.

When evaluating the suitability of onshore and offshore storage sites, we use two independent criteria systems because of differences in technical schemes and data sets. Techno-economic analyses for onshore and offshore storage are also conducted independently. Additionally, all analyses are grounded in the current understanding of large-scale carbon storage technologies, database resources, and the analytical algorithm system. It should be noted that this assessment serves only as a reference for macro-assessment and preliminary feasibility studies and cannot be used as a basis for screening specific sites and project design.

2.1. Technical Schemes for Large-Scale CO_2 Geological Storage

The performance evaluation of CO_2 geological storage depends on multiple factors, including various CO_2 trapping mechanisms and the spatial distribution of sedimentary facies, structures, lithologies, and related properties. Meanwhile, regional hydrodynamic conditions and the technical design of the CO_2 storage system impact site performance.[33−35]

Among these factors, the technical scheme for the CO_2 storage project can be meticulously designed to guarantee successful execution and performance. The key characteristics of large-scale storage include a near-mass-balance strategy, the use of multiple CO_2 injection and geofluid production wells, and the use of numerous geological strata to maximize underground space and mitigate geological risk. These other technical features include well-field design, well structure, multiple-layer optimization, and risk mitigation strategies, all of which are crucial for the safe storage of CO_2 and the maximization of underground space utilization. Subsequently, key site performance can be evaluated using hypothesized technical schemes and relevant algorithms for storage capacity, injectivity, site suitability, and economic evaluation.

2.2. Algorithms of Site Performance Evaluation

Based on previously established site assessment methods, site performance assessment can be conducted using GIS-based raster analysis or numerical simulation. The relevant GIS data are rasterized with a 1 km × 1 km spatial resolution. Subsequently, algorithms for site suitability, capacity, economic evaluation, and source-sink pairing are used to analyze and summarize the raster data set, yielding outcomes of "hierarchical capacities-costs-geospatial matching".

2.2.1. Sites Suitability. Any potential storage site must meet numerous criteria to ensure the effectiveness, safety, and economic feasibility of storage operations.[27,36] The site suitability assessment of CO_2 geological storage should consider multiple indices across three key aspects: (1) storage site performance (storage capacity, injection, and techno-economic performance); (2) storage safety or risk management; and (3) social-economic and other constraints (economic, legal, and environmental factors). A systematic classification of CO_2 storage indices is proposed and presented in Supporting Information Tables S2 and S3. This classification integrates geological performance, safety, and socio-environmental constraints, enabling high-resolution site suitability assessments for onshore and offshore aquifer sites. We used multisource data assimilation to normalize and classify assessment indices for geological site suitability. Then, we generated individual and summary index layers using GIS spatial analysis algorithms to conduct a high-resolution site-suitability assessment. This process helps analyze key factors affecting geological storage suitability and their influence levels for scientific site selection.[37]

2.2.2. Capacity for Storage Sites. The mathematical theories, evaluation procedures, and data requirements for the above-capacity algorithms differ substantially. The existing algorithms and tools for storage capacity estimation can be categorized into several major classes, including empirical methods, semianalytical methods, reduced-order methods, numerical simulations, and other methods. Most methodologies rely on the theoretical and geo-cellular volumes of storage reservoirs, integrating specific storage mechanisms over defined periods, spanning from the commencement of CO_2 injection to its termination or the final state of the injected CO_2.[38] The precise and detailed comparisons of existing algorithms depend heavily on the quality of the data set (in terms of various types and resolution).[39−42] Based on data quality, the algorithm should be chosen from empirical or statistical algorithms, analytical or analytic methods, reduced-order numerical simulation, numerical simulation, general artificial intelligence, and hybrid algorithms.[43−48] More detailed information on capacity algorithms can be found in the algorithm realization section of the Supporting Information.

2.2.3. CO_2 Emission Sources. Carbon emission source assessment involves conducting investigations and analyses of carbon-emitting enterprises, such as the coal-fired power, steel, cement, and chemical industries. It aims to understand: (1) basic information on the enterprise, including its location, products, output, and processes; (2) energy consumption of coal, oil, and gas, water, electricity, etc.; (3) carbon emission situations, such as carbon emission intensity, main emission links and concentrations, and emission amounts. A carbon-emission source database is established, and enterprise screening indicators are developed to assess whether carbon-emitting enterprises are suitable for implementing CCS projects. Different regions and industries need to develop screening criteria for their CO_2 sources. More detailed information about P10 and P90 results can be found in the capacity algorithms in the Supporting Information.

2.2.4. Techno-Economic Assessment of Full-Chain CCS Project. The costs of CO_2 storage operations are critically contingent upon an integrated set of strategies.[9,49] Techno-economic models, which incorporate algorithms, are composed of two main components: technical and economic models. Numerous economic models of onshore and offshore CO_2 geological storage have been developed globally.[50−57] An algorithm considering a broader range of technical schemes and economic parameters can yield cost estimates with higher resolution and reduced uncertainty. This study employs a technical-economic module at the prefeasibility scale for assessment. The economic viability of the storage site depends on the site performance assessment and the engineering implementation process. This encompasses factors such as capacity and injectivity, the type of offshore platform, the well pattern and layout, injection strategies, and cost parameters. In terms of technical details, economic models can be categorized into empirical, statistical, budgetary, and accounting models. More detailed parameters and assessments can be found in the techno-economic models in Supporting Information (key parameters shown in Tables S5−S10). All costs and revenues are escalated to 2022 USD.

2.2.5. Source-Sink Pairing Assessment. Source-sink pairing algorithms aim to pair CO_2 emission sources (e.g., power plants, industrial facilities) with suitable geological sinks

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

(e.g., depleted oil reservoirs, saline aquifers, chemical utilization) while minimizing costs and risks 62, 63 The source-sink pairing model is developed based on different models of integrated CCS projects. The model includes point-to-point source-sink pairing (P2P) and multiple-source-to-multiple-sink pairing (M2M) scenarios.[58] Pipeline optimization algorithms were employed to identify suitable routes that considered technical features, socioeconomic factors, environmental impact, and other relevant factors. The costs of each source-sink pairing or of integrated CO_2 sources and sinks within the search radius can be assessed using reservoir properties, regional coefficients, technical and economic parameters, and the techno-economic model. In this assessment, all modeled CO_2 sources are assumed to promptly initiate CCS retrofitting and commence operations at their assumed capacity levels. This assumption focuses on defining the key attributes of matched CCS projects to understand the commercialization potential of CCS technology across China. More detailed information about the full description and parameter settings can be found in the Supporting Information.

2.3. Data Collection and Integration

The data supporting the evaluation method includes various data types and levels of precision, as well as data transition or assimilation methods, which jointly support the operation of the evaluation model.

2.3.1. Data Set. Geological surveys generate data sets taxonomically divided into subsurface and surface domains, each characterized by distinct spatial−temporal resolutions and scientific utilities.[59,60] Subsurface data sets exhibit hierarchical organization across nested scales: (1) basin-scale frameworks integrate stratigraphic architecture, sedimentary facies, lithophysical property distributions, and geodetic observations to construct reservoir-caprock system models; (2) regional-scale attributes synthesize reservoir-seal integrity, boundary system typology (open/semiopen/closed), fault activity regimes, and facies continuity to constrain basin-scale fluid migration; (3) site-scale data sets combine petrophysical measurements, borehole lithology, 2D/3D seismic data, and high-permeability conduit delineation (sparse in oil fields and CO_2 storage projects).

Surface-derived data sets, critical for anthropogenic-environmental-social-economic analyses, include: (1) surface geological archives (outcrop lithology, stratigraphic contouring, structural morphology); (2) geospatial-geomorphic data sets (digital elevation models, river and lake system, landform classifications, climatic parameters like precipitation/insolation/wind regimes, and vegetation indices via satellite remote sensing); (3) Socio-environmental metrics (CO_2 emission hotspots, transportation networks, urban clusters, industrial zones, protected areas, and economic indicators), enabling sustainability-focused spatial planning.

Data sources include institutional repositories (Chinese Academy of Sciences, Ministry of Natural Resources, National Basic Information Center), requiring continuous updating to support zone- and site-scale research. Subsurface geological/resource data, as well as surface-derived data sets (both geological and nongeological), are detailed in the Supporting Information Table S13. Techno-economic parameters incorporate current petroleum industry and CO_2 storage project metrics, spanning site exploration, drilling, oil and gas fields, and pipeline operations.[61] This hierarchical data set architecture, underpinned by rigorous scale-transition methods, forms a foundational toolkit for addressing complex challenges in the Earth system, particularly in energy, environmental, and engineering geosciences.

2.3.2. Data Implementation and Quality Definition. The data implementation framework establishes hierarchical fusion strategies spanning core-scale to regional-scale data sets, with subsurface data resolutions and types detailed in Table 1.

Table 1. Hierarchical Types and Resolutions of Zone-Scale Geological Data[59]

exploration stage	areal resolution of GIS data	spacing of well data	resolution levels
CO_2 injection (a)	≤1:1 million	at least one well per 25 km^2 or average well spacing <5 km	site-level
detailed Characterization (b)	≤1:2.5 million	at least one well per 100 km^2 or average well spacing <10 km	zone-level
investigation (c)	≤1:5 million	at least one well per 400 km^2 or average well spacing <20 km	sub-basin level
general survey (d)	≤1:10 million	at least one well per 2500 km^2 or average well spacing <50 km	basin-level

Site-scale characterizations (stratigraphic sequences, well logs), regional geological data sets (1:5 M−1:250 K-scale maps integrating stratigraphic thickness, geological age, and isotopic data), and remote sensing imagery are integrated via upscaling, downscaling, and scale-similarity processing. This enables same-scale analysis and error quantification across hierarchical layers, culminating in integration of multiscale, multispatial geological and surface data sets using web-based GIS and matrix data systems. Advanced integration frameworks leverage data fusion methods to synthesize multiresolution data sets into a uniform spatial scale (about 1 km × 1 km or 0.01° × 0.01° resolution in this study), forming the basis for capacity assessment, techno-economic evaluation, site screening, and source-sink pairing.

The high-resolution (1 km × 1 km) spatial analysis presented in this study is based on geological source data with a much coarser resolution (well spacing at 20−50 km). While we employed a data interpolation tool and other downscaling approaches to generate the high-resolution grid, it is crucial to acknowledge the inherent uncertainties in this process. The fidelity of the resulting model is fundamentally constrained by the quality and density of the underlying input data. Although the downscaling method enhances visualization precision and introduces spatial variability consistent with the source data, it cannot resolve subgrid geological heterogeneities (e.g., small-scale faults, channels, or permeability barriers) that are not captured by the sparse well data. Therefore, the model represents the best possible interpretation given the available data, and the results should be interpreted with the understanding that the resolution of fine-scale features is limited by the original data density. These methods are suitable for evaluating the general technical economy and suitability of sites at the zone scale, but a more refined assessment of site-specific heterogeneities would require denser, higher-quality input data, such as 3D seismic surveys or additional well logs.

2.3.3. Evaluation Framework. Building on the multi-dimensional data set, a four-pillar evaluation system is constructed, forming a zone-scale framework within the Resources-Engineering-Economics-Policy Integrated Assessment Framework for CCS. This platform visualizes zone-level suitability indices, hierarchical capacities, and levelized costs,

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

enabling stakeholders to screen storage sites and design CCS projects under different policy scenarios. This tool also addresses a critical need for actionable, spatially explicit analytics in global road-mapping for carbon capture, utilization, and storage (CCS).

3. RESULTS AND DISCUSSION

Utilizing the database and evaluation framework, the CO_2 sources, site suitability, hierarchical capacities, techno-economic features, and source-sink pairing of CO_2 streams with onshore and offshore aquifer sites were assessed.

3.1. CO_2 Emission in Coal-Based Energy and Industry in China

Sector-specific data sets were systematically compiled from China's coal-fired power plants, crude steel production facilities, cement manufacturing units, and coal chemical industries in 2022. The integrated database encompasses geographic locations, production capacities, actual outputs, key industrial processes, and resource consumption patterns (including fuel, water, steam, and electricity) across these energy-intensive sectors. The inventory of CO_2 emissions from coal-intensive sectors in this evaluation is presented in Table 2.

Table 2. Inventory of CO_2 Emissions from Coal-Intensive Sectors in This Evaluation

item	source number	CO_2 stream (Mt/a)	high-concentration CO_2 (Mt/a)	CO_2 quality
coal-fired power	1133	4658.2		12−15% Vol
crude steel	280	1763.4		20−25% Vol
coal chemical	312	392.7	226.0	12−15% and >85% Vol
cement	2948	2099.9		25−30% Vol
summary	4673	8914.1	226.0	

CO_2 emissions from coal-fired power plants were obtained from the China Electricity Council (CEC), calculated based on annual electricity generation and technology-specific emission factors (e.g., ultrasupercritical and supercritical units). Emissions from blast furnace ironmaking, sintering, and coke-oven coking processes dominated crude steel production. Three product categories (cement, clinker, and mixed materials) were assigned differentiated emission factors in the cement production process. The modern coal chemical processes involve process-related CO_2 emissions from chemical reactions (e.g., coal gasification/liquefaction) and conventional CO_2 emissions from ancillary operations. The CO_2 capture system design is based on CO_2 concentration-stratified optimization. China's combined annual CO_2 emissions from these four sectors reached 8.9 Gt in 2022.[12,15,36] The spatial distribution of CO_2 sources is illustrated in Figure 4. Under an 85% net capture rate scenario (representing the current technological optimum), 7.6 Gt CO_2/year could be theoretically captured. Potential CO_2 emission double-counting exists between coal-fired and captive power plants (supplying cement, steel, and coal chemical sectors), which should be noted to avoid overestimation. Small-scale captive plants (emitting 320 Mt/a) and excluded CO_2 sources (<500 kt/year) have negligible impacts on coal power's total 4.34 Gt/a emissions; these limitations are noted.

3.2. Basic Geological Setting

China's sedimentary basins exhibit distinct structural and stratigraphic characteristics across terrestrial and marine realms, shaped by diverse tectonic histories and depositional environments. The large sedimentary basins in China and its adjacent regions are shown in Figure 4.

China's basins—extensional in the east, compressional in the west, and rifted-marine offshore—offer diverse carbon storage potentials. Onshore basins vary in reservoir properties and caprock stability. In contrast, offshore basins feature thick, continuous sealing strata, which are critical for optimizing onshore-offshore sequestration strategies to meet decarbonization goals efficiently.

Eastern onshore basins, such as the Songliao and Bohai Bay basins, are fault-depression/extensional types formed under Mesozoic−Cenozoic tension, characterized by thick sequences of Mesozoic and Cenozoic clastics. Reservoirs exhibit moderate porosity and permeability due to diagenesis, posing challenges for large-scale carbon storage, despite the presence of favorable structural traps for hydrocarbons. Central basins (Ordos, Sichuan), governed by compressional-shear forces, host Paleozoic−Mesozoic sediments, including low-porosity and low-permeability (5−15% and 0.1−10 mD) sandstones. While challenging for conventional oil and gas recovery, their stable cratonic settings with sealed strata benefit storage safety. Western basins (Tarim and Junggar), formed by compressional tectonics, span Paleozoic to Cenozoic deposition and contain up to 10,000 m of sedimentary rocks. Target strata with suitable sandstones (12−20% porosity) that benefit from robust structural traps and regional seals, making them promising for CO_2 sequestration. Offshore basins (East China Sea Shelf, Bohai, Pear River Mouth Basins et al.) formed during the Late Mesozoic−Cenozoic, via early continental rifting and late marine depression. Neogene-Quaternary marine sediments exceed 8000 m in thickness, creating clastic-carbonate reservoir-caprock associations (e.g., high-porosity continental sandstones sealed by marine muds) to favor carbon storage.[62]

3.3. Site Suitability

The site assessment methodology and fundamental data were utilized to conduct the site suitability assessment for CO_2 storage in saline aquifers. Figure 5 shows the assessment results of site suitability for CO_2 saline aquifer storage in the sedimentary basins of China and adjacent areas. Generally, saline aquifers in sedimentary basins in China and neighboring regions exhibit relatively high site suitability. The assessment outcomes reveal numerous suitable sites for saline aquifer storage. Among the offshore basins, the Bohai Basin, the Yellow Sea Basin, the East China Shelf Basin, the Pearl River Mouth Basin, and the Yinggehai Basin stand out for their relatively high site suitability for CO_2 storage. These basins boast abundant exploration data, large storage capacity density, good injectivity, high safety, and low risk. Combined with onshore aquifer sites, China's total theoretical and effective capacities are estimated to be approximately 2.5 and 1.5 trillion tons, respectively. Additionally, onshore and offshore aquifer storage costs typically fall below $10 and $30 per Ton, respectively. Storage sites with higher suitability can be strategically combined with emissions sources from coastal cities to initiate demonstration projects, thereby laying the foundation for technological advancement of offshore CO_2 storage in saline aquifers.

3.4. Hierarchical Capacities

CO_2 storage capacity assessments are conducted using databases specific to each sedimentary basin, and all basins are then classified according to administrative and tectonic boundaries.

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

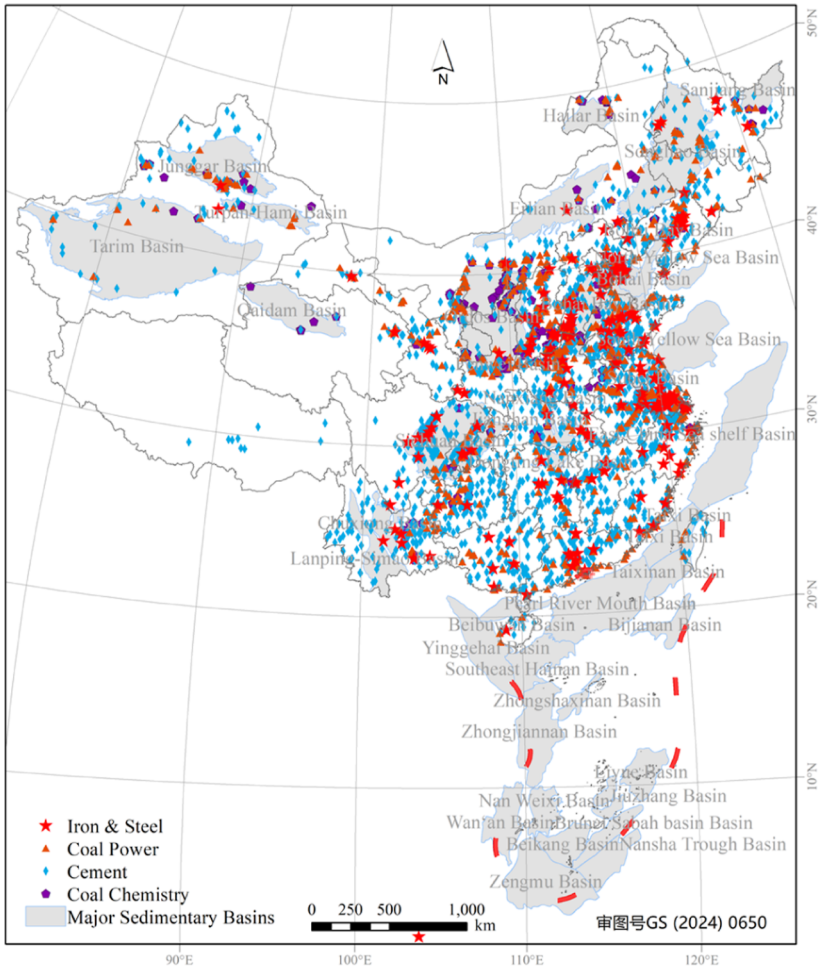


Figure 4. Distribution of CO_2 sources and large sedimentary basins in China and adjacent regions.

The resulting theoretical and effective storage capacities are approximately 2.5 and 1.5 trillion tons, respectively.

Figure 6 illustrates the effective capacity of each structure, represented by storage capacity density, which correlates with the structure's area. The storage capacity density ranges from 0.01 to 1.35 Mt/km^2. Offshore basins exhibiting relatively large storage capacity include the Zengmu Basin, East China Sea Basin, Pearl River Mouth Basin, and Yinggehai Basin. These basins are characterized by their expansive areas, substantial thicknesses of Cenozoic sedimentary rocks, and favorable physical properties, which contribute to their significant storage capacity and suitability. Onshore regions also have high-capacity density, including the Tarim, Junggar, Qaidam, Ordos, Songliao, and Bohai Bay Basin (onshore). They also benefit from the expansive basin area, substantial thickness of sedimentary rocks, and proper physical properties. These basins are characterized by their large basin area, substantial sedimentary rock thickness, and favorable physical properties, which contribute to their significant storage capacity and suitability. The hierarchical capacities of CO_2 aquifer storage in each sedimentary basin are summarized in Table 3. The theoretical storage capacity ranges from 1.25 to 3.76 trillion tons (P10−P90), with the P50 effective capacity on the order of 1.5 trillion tons. The matched capacities range from 4.2 to 6.1 Gt/a of captured CO_2 (7.6 Gt/a) from the coal and energy sectors, with pairing distances ranging from 150 km to 800 km, annually. More detailed results for each basin are shown in Supporting Information Table S14.

3.5. Storage Cost

Figure 7 illustrates the average levelized cost of CO_2 geological storage in China and its adjacent regions. Basins such as the Bohai Bay Basin, East China Sea Basin, Pearl River Mouth Basin, Yinggehai Basin, Zengmu Basin, and some basins in the South China Sea exhibit favorable injectivity. This is attributed to the thick Cenozoic sedimentary rocks and their good physical properties in these regions. The risk and cost associated with saline aquifer storage are low in areas with good injectivity. The costs of most offshore storage are generally below 60 USD/t. Offshore sites range from 10 to 60 USD/t. In contrast, onshore storage ranges from 3 to 10 USD/t, driven by factors such as offshore platforms, related offshore infrastructure, heterogeneous geological features, capacity intensity, and injectivity. Large-scale CO_2 injections can lead to high-pressure build-up, limiting storage capacity and potentially increasing long-term risks. Therefore, during site performance evaluation, it is essential to establish a comprehensive injection or storage strategy that considers the layout of the well field, the number of wells to be drilled, and the design of pressure-control (water-recovery) wells.

3.6. CCS Cost Curves and Various Heterogeneities

Accurate assessment of source-sink pairing and proximity for CCS in key regions is critical to accelerate large-scale deployment of CCS projects and hubs. Analysis shows that >99% of CO_2 sources (coal power, steel, cement, and coal

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

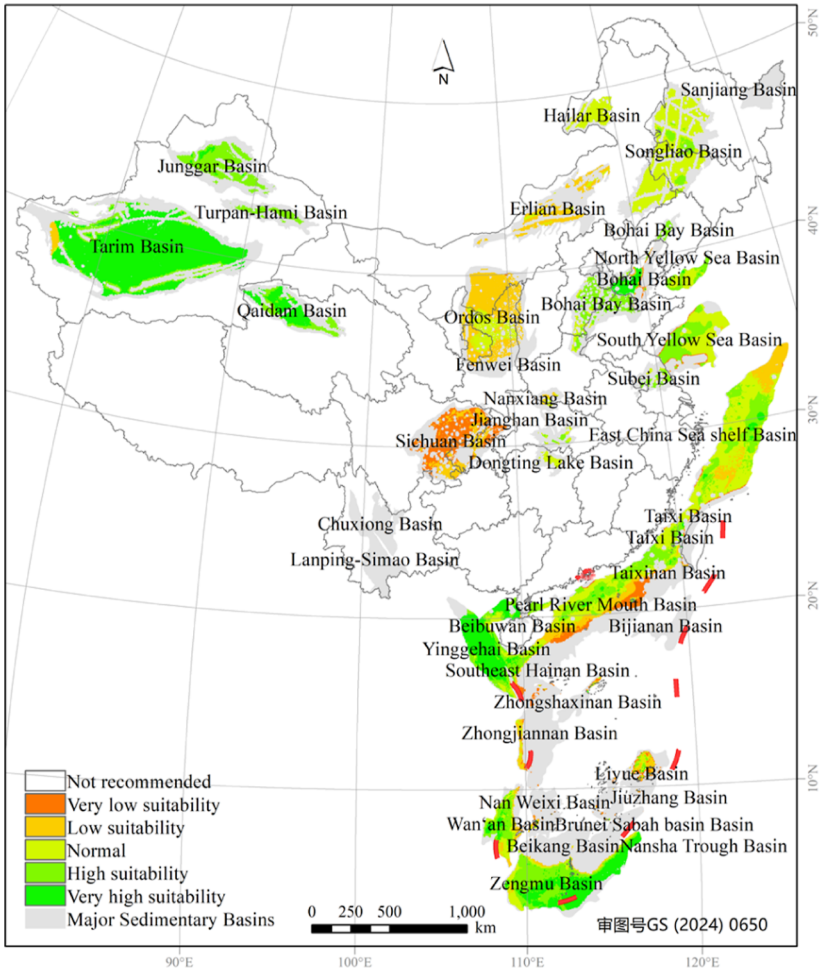


Figure 5. Spatial distribution of site suitability in China and adjacent areas.

chemical sectors) in China can be paired with suitable onshore or offshore storage within 800 km, though mitigation costs exhibit strong distance and spatial dependency. Cost curves for CCS mitigation potential (Figure 8) reveal that at 85% capture rate, achievable abatement rises from 4.2 Gt/a (150 km pairing) to 6.0 Gt/a (800 km) with full-chain levelized costs <90 USD/t, covering 59−79% of annual CO_2 emissions captured from target industries. Slight upticks in avoidance and levelized cost curves reflect CCS project-induced emissions and reliance on coal-fired energy. The plateaus in levelized costs and carbon abatement cost curves stem from transitions between onshore/offshore storage systems and sedimentary basin heterogeneities. Onshore reservoirs offer lower infrastructure and operational costs, making them economically preferable for an increasing number of sources as pairing distances expand, thus shortening the cost plateau. Beyond critical thresholds (e.g., 300 km), distance-dependent transport logistics dominate cost gradients. This is driven by high single-pipeline costs that reduce onshore viability, particularly in high-GDP regions with steep land acquisition costs, thereby incentivizing phased offshore integration.

Regional heterogeneity in costs arises from distributions of storage costs (Figure 9) and gaps in transport infrastructure, including urban, hydrological, railway, and conservation area impacts. This necessitates backbone pipeline and cluster strategies for economic deployment. Globally, China's CCS cost curves are low: full-process costs for >69% of CO_2 capture lie in 30−60 USD/t, and >79% (about 4/5th) of cases cost <90 USD/t. However, a higher fraction of projects exceeds 60 USD/t than in other nations,[37,57,63] attributed to source-sink imbalances necessitating long-distance transportation and offshore storage. Future policies must prioritize long-distance transport and cluster development to align with China's unique CCS landscape.

The site suitability levels, storage capacities, levelized costs, CCS project costs, and regional characteristics presented in this analysis were compared with the prefeasibility study results of multiple large-scale CCS demonstration projects in China, including the Ordos coal-to-liquids, Yulin coal chemical, and CNOOC Enping projects, as well as the Bohai Bay project. Under the P50 probability level, the site suitability, capacity, and costs are generally consistent with the actual full-process project costs.

3.6.1. Cost Curve Characteristics: Administrative and Basin Perspectives. When categorized by storage sites (Figure 9), cost curves are shaped by site-specific attributes, proximity to emission sources, and source-sink dynamics. Bohai Bay, Ordos, Subei, Songliao, Junggar, Jianghan-Dongting, and East China Sea Shelf Basins showcase CCS curves with substantial emission reductions and gradual cost transitions. Marine basins incur significantly higher abatement costs than their continental counterparts due to multistep profiles, which arise from heterogeneous storage capacities across intrabasin zones, leading to tiered transportation and storage expenses.

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

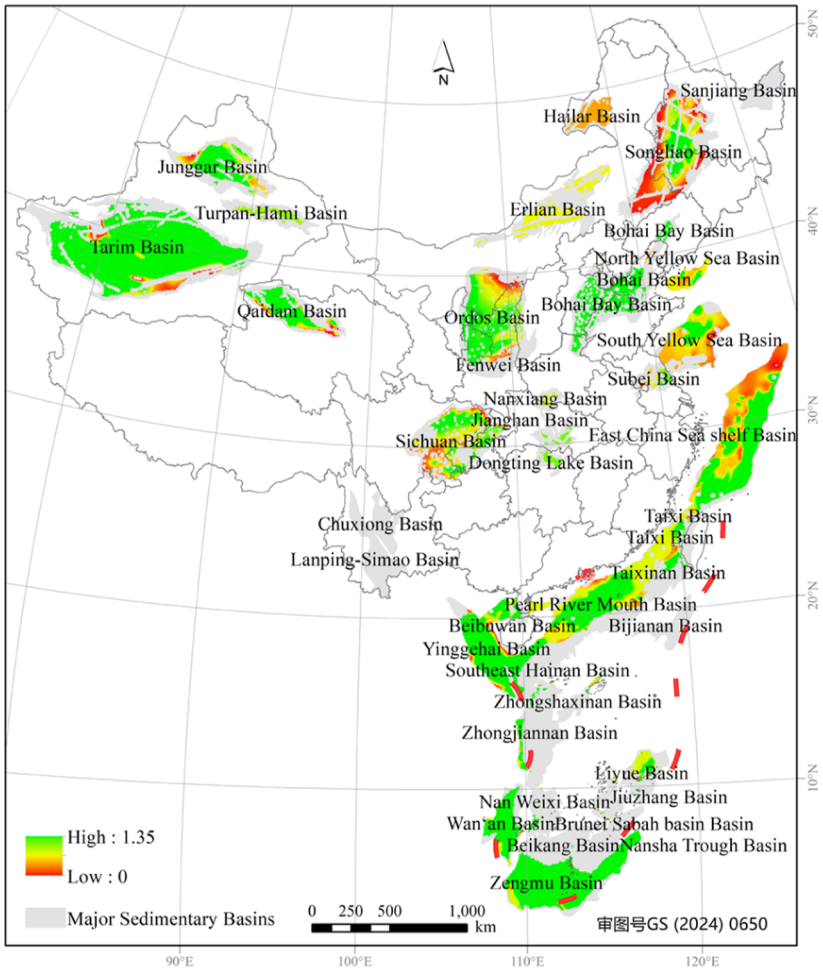


Figure 6. Effective capacity density in China and adjacent areas (units: Mt/km^2).

Table 3. Hierarchical Capacities of CO_2 Geological Storage in China (P50)

basins	theoretical capacity	effective capacity	matched capacity with 150 km	matched capacity with 300 km	matched capacity with 800 km
onshore Basins	1305.7 Gt	756.2 Gt	3757.7 Mt/a	4993.1 Mt/a	5559.2 Mt/a
offshore Basins	1201.6 Gt	743.6 Gt	453.3 Mt/a	476.2 Mt/a	548.2 Mt/a
total	2507.3 Gt	1499.8 Gt	4211.0 Mt/a	5469.3 Mt/a	6107.4 Mt/a

When the cost curve with 300 km matching distance is categorized by administrative regions (shown in Figure 10), Cost curves exhibit distinct patterns—smooth transitions or stepwise features—across administrative areas, governed by source distributions (discrete point source or centralized hub in industry parks), zonal properties of storage site, and source-sink pairing efficiencies. Provinces like Shandong and Hebei exhibit smooth cost transitions due to centralized storage within the Bohai Bay Basin, where gradual changes in transport distances and storage site properties minimize cost fluctuations. In contrast, multistep features emerge in regions. For example, the cost curves of Inner Mongolia and Xinjiang span multiple sedimentary basins and blocks. At the same time, Jiangsu's coverage includes subregions within the Subei Basin and South Yellow Sea Basin. This is due to disparities in storage site selection across zones by emitters, which introduce discrete jumps in total costs.

The CCS emission reduction curves vary significantly across administrative regions, resulting in notable administrative heterogeneities in the economics and contributions of traditional energy industries to CCS technologies. The scale and cost-effectiveness of future emissions reductions are closely linked to each region's resource endowments, industrial foundations, and emission-reduction policy planning. Notably, cross-administrative regional carbon mitigation strategies significantly shape carbon neutrality pathways within individual areas, underscoring the crucial role of interjurisdictional coordination in national decarbonization trajectories.

3.6.2. Regional and Sectoral Heterogeneity. Heterogeneity in geological storage capacity and matching costs for CCS emerges across basins, regions, industries, and administrative boundaries, shaped by physical attributes, economic landscapes, and policy frameworks. Basin-wise, continental basins (e.g., Bohai Bay, Ordos) exhibit gradual cost transitions with high emission reduction potential. In contrast, marine basins (e.g., East China Sea Shelf) incur stepwise higher costs due to intrabasin storage heterogeneity, which drives tiered transport and storage expenses. Regionally, low-GDP coal-rich zones (Xinjiang, Inner Mongolia) offer cost-competitive onshore sequestration via favorable subsurface properties and

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

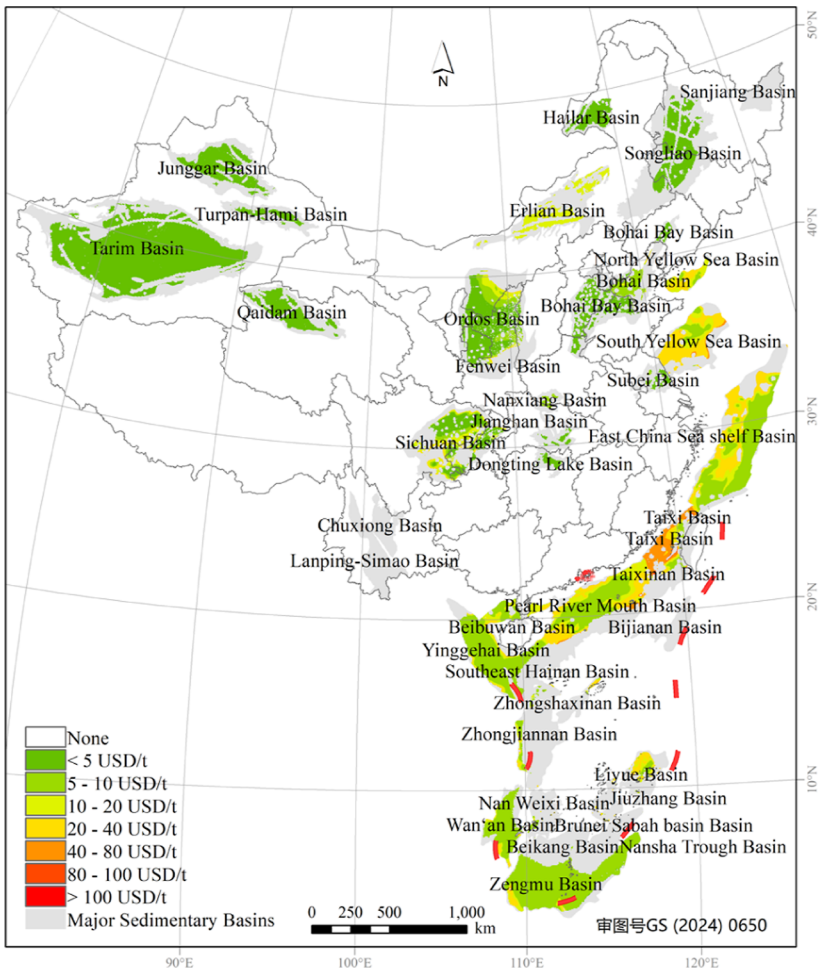


Figure 7. Average levelized cost of large-scale CO_2 geological storage in China and adjacent sea areas.

low land/energy costs. At the same time, high-GDP coastals leverage marine basins to halve abatement costs through proximal source-sink pairing, despite elevated offshore infrastructure investments. Sectoral disparities stem from CO_2 stream characteristics: coal chemical industries benefit from high-concentration streams, reducing capture costs; coal-fired power/steel plants adopt pithead models for moderate costs with proximal storage; cement sectors face the highest costs due to fragmented scales and geographic isolation. Administratively, cost curves vary from smooth transitions in provinces with centralized basin storage (Shandong in Bohai Bay) to stepwise profiles in multibasin regions (Inner Mongolia), reflecting source-sink pairing efficiencies and intrabasin storage heterogeneities. Such heterogeneities necessitate spatially tailored policies—cross-regional carbon corridors and sector-industry integration—to align CCS deployment with resource endowments and optimize decarbonization costs.

Regional CCS coalitions classified by source-sink pairing form independent CO_2 reduction hubs, exhibiting marked cost and emission reduction scale heterogeneities. Twelve coalitions are proposed based on sequestration priority and resource endowment, with Junggar-Tuha, Tarim, Qaidam, Ordos, Songliao, Jianghan-Dongting, Erlian, Subei-onshore/South Yellow Sea-offshore, Bohai Bay-onshore/Bohai-offshore, East China Sea Shelf, Pearl River Mouth, and Beibuwan identified as prime CCS cluster candidates. Distinct zone- and site-level storage evaluations inform optimal site selection and cross-sector/regional collaboration modules, with tailored strategies: coastal regions like Subei prioritize integrated land-sea storage due to limited onshore capacity. Large-scale, low-cost "cross-regional carbon corridors" could facilitate CCS deployment and industrial relocation to lagging regions.

Cross-industry and regions collaboration accelerates CCS scaling, but heterogeneities necessitate phased strategies: short-term enterprise alliances (e.g., China Energy, Huaneng Group, China Offshore Oil Company et al.) for pipeline sharing; midterm provincial-level carbon mitigation agreements; long-term statutory "carbon corridors" integrated into cross-regions and national spatial planning. This roadmap transitions from enterprise-led infrastructure sharing to policy-governed coordination to address scaling challenges. The decarbonization-industrial revitalization dual strategy uses 12 cross-provincial coalitions as hubs:

- Cost synergy pools regional industries (e.g., high-efficiency manufacturing) and low-carbon energy (solar, wind) to reduce CCS costs via cross-sectoral integration of CO_2, low-cost power, and industrial byproducts.
- Technical synergy bridges onshore-offshore divides (e.g., Subei/South Yellow Sea), combining high-efficiency onshore sequestration with low-cost marine storage.
- Policy-driven regional synergy establishes "carbon corridors" to relocate emission-intensive industries to the Three Norths regions with abundant onshore storage,

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

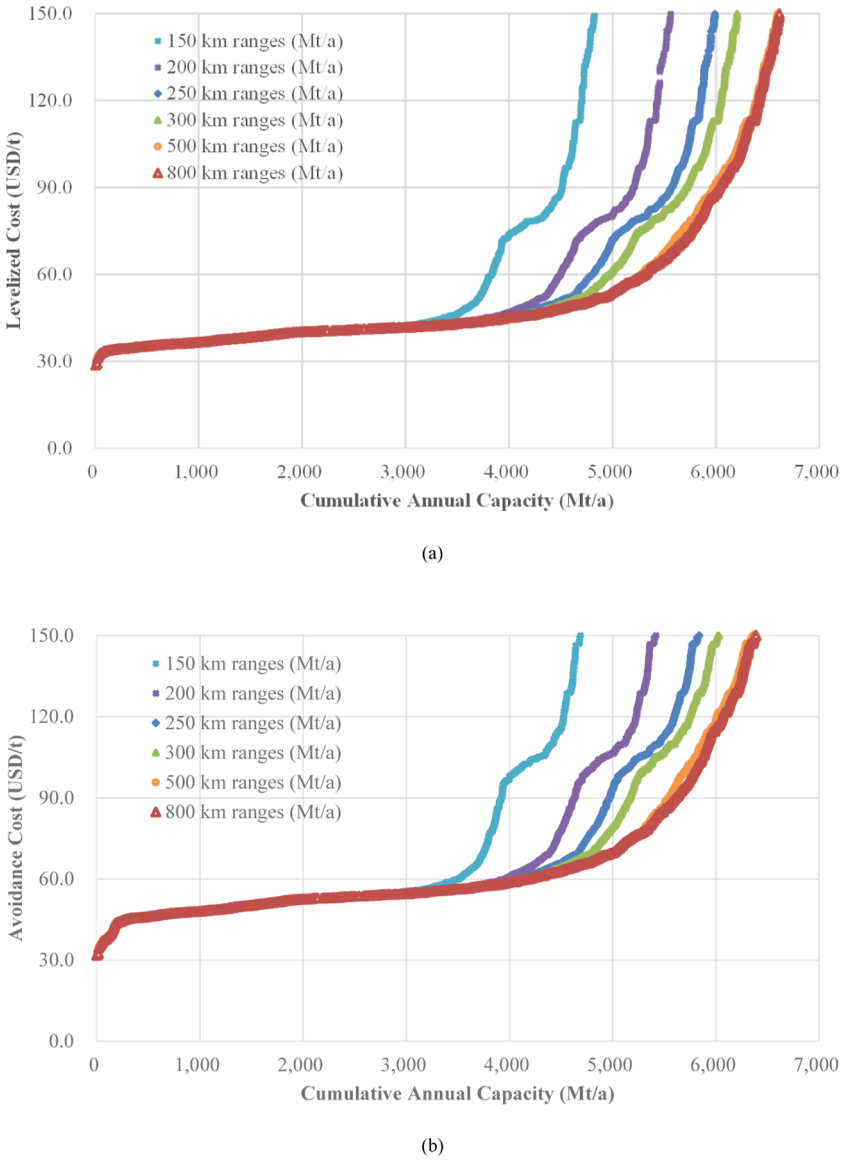


Figure 8. Cost curves of CCS mitigation potential in China for various pairing distances.

rebalancing growth between developed and underdeveloped areas.

This mechanism ties decarbonization to industrial revival via cost control, technical integration, and policy-enabled regional flows, positioning CCS as a dual-benefit driver for China's low-carbon transition.

3.6.3. Competition from a Global Perspective. The product offers economic viability, with lower overall costs and greater competitiveness compared to counterparts in the U.S., Central Asia, and Europe. This competitiveness stems from synergies in infrastructure deployment, economies of scale, and integrated industrial networks. Concurrently, it highlights the resilience of China's traditional coal-based industries under the carbon neutrality agenda, illustrating how these industries can leverage carbon capture and storage (CCS) technologies to advance decarbonization while mitigating stranded asset risks, ensuring a smooth low-carbon transition, and complying with regulatory requirements. The country's unique position in balancing low-cost CCS implementation with industrial transition highlights its potential to shape global carbon management strategies.

3.7. Limitations and Future Work

This study presents a novel zone-scale methodological framework at 1 km grid resolution, complementing existing site-scale tools. By integrating capacity-suitability-economic triads through geospatial analytics, it surpasses conventional single-reservoir models, enabling national CCS infrastructure planning with strategic hub identification, such as the Ordos Basin's $8 USD/t geological storage cost. This systemic approach bridges scale and resolution gaps in carbon management, shifting CCS deployment from isolated projects to integrated networks.

However, limitations exist in both methodology and assessment. Potential double-counting of coal-based power emissions across thermal and captive power plants may overestimate China's CO_2 emissions in four sectors, requiring careful consideration. Uncertainty analysis of the capacity-suitability-economics-cost curve is constrained by limited parameter uncertainty ranges in existing geological data sets and by high computational costs for multidimensional analysis. Current

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

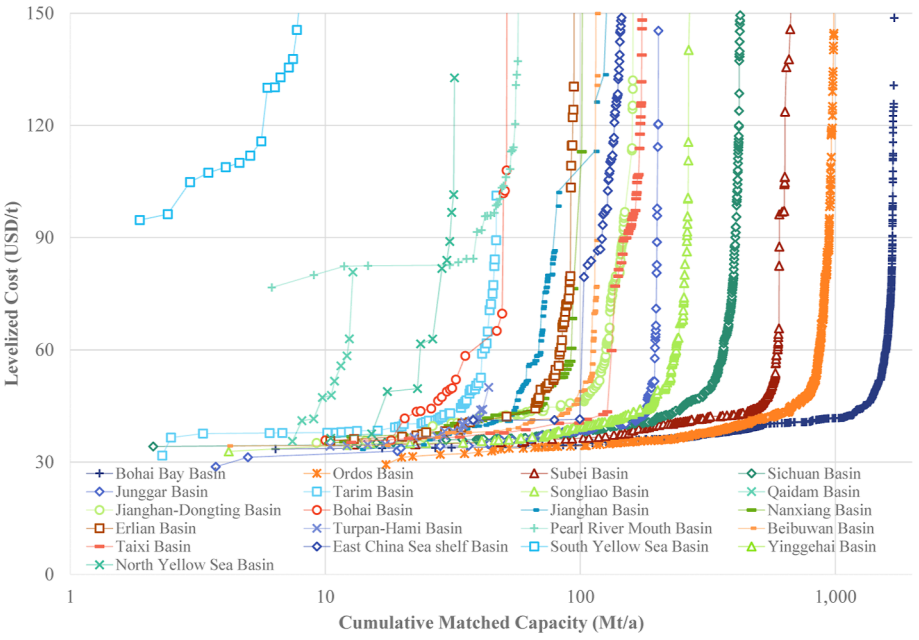


Figure 9. Carbon abatement cost curves for various matched basins with source-sink pairing conditions in China (300 km).

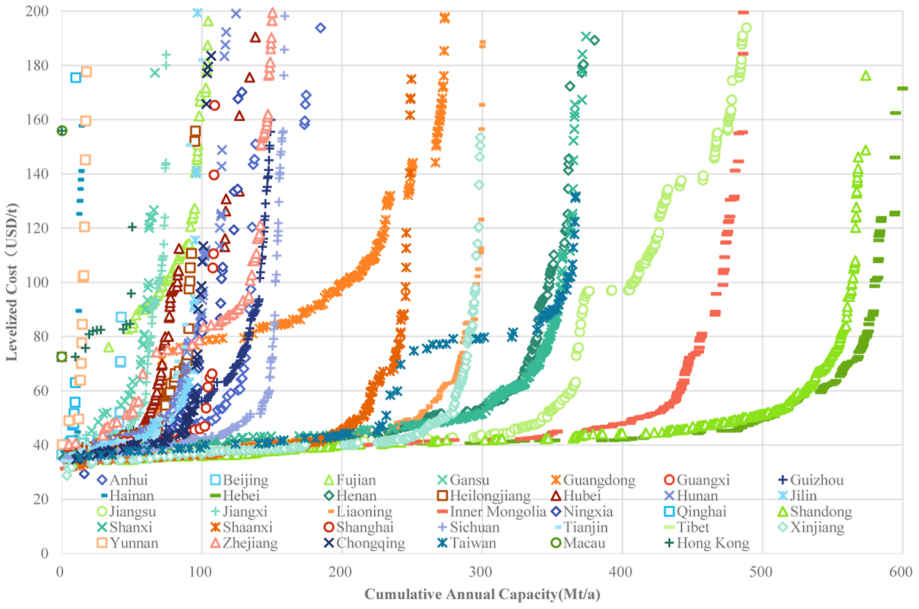


Figure 10. Cost curves for various provinces with source-sink pairing constraints in China (300 km).

results primarily rely on P50 values, with only capacity providing P10−P90 ranges, as the public focuses on P50-level triad outcomes.

This paper primarily focuses on saline aquifer storage. Even if EOR generates profits, its trading price still needs to be set at the standard price (mainly derived from saline aquifer storage with the most supply). Meanwhile, this paper conducts assessments targeting large-scale saline aquifer storage, thus excluding oilfield-related content.

All quantitative source-sink matching results rely on P50 median estimates; actual outcomes may be affected by cost uncertainties, primarily stemming from transportation and reservoir properties 25. The findings serve only for strategic planning and policy support rather than for final project design, and future work will enhance robustness through a comprehensive P10−P90 full-range uncertainty analysis.

Additionally, even with mitigated CCS operational emissions, coal-based industries and power systems cannot achieve 100% net CO_2 mitigation solely through CCS. This necessitates integration with complementary technologies like biomass energy, direct air capture with geological storage, and low-carbon power sources. Future improvements will involve a comprehensive uncertainty analysis across P10−P90 ranges for capacity, suitability, and economic contribution. Additionally, refining technical parameters, assessment indices, and weights will be pursued as engineering experience, data quality, and techno-economic understanding advance, potentially significantly enhancing the accuracy of results.

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

4. IMPLICATIONS

This study addresses the spatial mismatch between CO_2 emission sources and geological storage sites in China by introducing a high-resolution geo-spatial techno-economic framework for carbon capture and storage. The framework integrates hierarchical capacity assessment, multicriteria site screening, techno-economic modeling, and source-sink pairing algorithms to optimize the deployment of onshore and offshore storage. Key findings reveal:

- China's theoretical onshore and offshore CO_2 storage capacities are nearly balanced (2.5 trillion tons total), with 1.5 trillion tons of effective capacity. Cost-surface modeling shows 64−75% of industrial emissions can be stored within 300 km at costs <90 USD/t.
- The ITEAM-CCS platform fuses multisource data (geological, emission, and project parameters) to enable spatially explicit CCS decision-making, filling a critical gap in global analytics.
- A first-of-its-kind zone-scale database covers 16 onshore basins and 15 offshore areas, quantifying capacities, costs, and source-sink pairing heterogeneities in different regions and sectors under diverse scenarios.
- Multidimensional analysis (1 km × 1 km resolution) uncovers regional-source-sink imbalances, prioritizes 12 cross-provincial CCS coalitions, and proposes "carbon corridors" to leverage land-ocean synergies, cross-regional and cross-industrial relocation for cost efficiency.

Implications:

This study provides a scalable blueprint for China's CCS deployment, systematically resolving spatial−temporal questions (5W2H) for large-scale CO_2 storage projects. The framework's granular insights into storage capacity, cost competitiveness, and policy-driven collaboration offer three key advancements:

- Scientific foundation for decarbonization: the geo-spatial techno-economic framework enables data-driven prioritization of CCS projects and advances the "tiered geological-economic assessment" concept, resolving the long-standing scale mismatch between emission inventories (plant- and process-level) and land-sea synthetic storage sites (granularity of km-level).
- Policy and industrial guidance: cross-regional/cross-industrial CCS coalitions and "carbon corridors" can mitigate source-sink mismatches to some extent, leveraging economies of scale and existing industrial clusters to reduce abatement costs. This dual strategy of decarbonization and industrial revitalization is particularly relevant for regions with carbon-intensive economies.
- Global replication for developing nations: the land-ocean integrated model and data-driven multiple-dimensional tools offer a transferable framework for Global South countries facing similar challenges, such as carbon-locked energy systems, uneven development, and complex source-sink geographies. By aligning geological endowments with industrial emission patterns, this approach can accelerate the adoption of low-cost CCS while supporting regional economic transitions.

This work underscores the critical role of geospatial analytics in bridging scientific understanding and practical CCS implementation, positioning China as a pioneer in integrating technical, economic, and policy dimensions for scalable carbon management. Meanwhile, it offers a replicable blueprint for Global South nations grappling with source-sink mismatches. The proposed "decarbonization−industrial revitalization" dual strategy offers a replicable template for Global South nations with (1) carbon-locked energy systems; (2) uneven regional development; (3) combined land-ocean storage endowments.

■ ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c14041.

SM1: Methods for the Zone-scale Source-Sink Pairing for CO_2 Capture and Geological Storage (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Ning Wei − *Institute of Rock and Soil Mechanics, National Key Laboratory of Rock and Soil Mechanics and Engineering Safety, Chinese Academy of Sciences, Wuhan, Hubei 430071, China; University of Chinese Academy of Sciences, Beijing 100049, China;* orcid.org/0000-0001-8763-9998; Email: nwei@whrsm.ac.cn

Authors

Keyao Lin − *Institute of Rock and Soil Mechanics, National Key Laboratory of Rock and Soil Mechanics and Engineering Safety, Chinese Academy of Sciences, Wuhan, Hubei 430071, China; University of Chinese Academy of Sciences, Beijing 100049, China*

Shengnan Liu − *Institute of Rock and Soil Mechanics, National Key Laboratory of Rock and Soil Mechanics and Engineering Safety, Chinese Academy of Sciences, Wuhan, Hubei 430071, China*

Dalin Jiang − *China Energy Group Economic and Technological Research Institute Co., Ltd, Beijing 102211, China*

Wendong Wang − *Institute of Rock and Soil Mechanics, National Key Laboratory of Rock and Soil Mechanics and Engineering Safety, Chinese Academy of Sciences, Wuhan, Hubei 430071, China*

Zunsheng Jiao − *University of Wyoming, Laramie, Wyoming 82071, United States*

Robert T. Dahowski − *Pacific Northwest National Laboratory, Richland, Washington 99354, United States*

Xiaochun Li − *Institute of Rock and Soil Mechanics, National Key Laboratory of Rock and Soil Mechanics and Engineering Safety, Chinese Academy of Sciences, Wuhan, Hubei 430071, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.est.5c14041

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

National Natural Science Foundation of China (NSFC) (No. 72243011) and China's National Key R&D Program (Grant no. 2023YFB4104200), as well as the collaboration from the U.S.-

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/acs.est.5c14041.pdf by guest on 25 October 2026

China Clean Energy Research Center (Grant no. 2019YFE0100100).

REFERENCES

(1) Prado, A.; Mac Dowell, N. The cost of permanent carbon dioxide removal. *Joule* **2023**, *7* (4), 700−712.

(2) Fan, J.-L.; Xiang, X.; Yao, Y.; Li, K.; Li, Z.; Wei, S.; Diao, Y.; Ju, Z.; Li, X.; Li, X.; Peng, B.; Ma, J.; Qi, S.; Zhang, X. Dataset of CO_2 geological storage potential and injection rate capacity in China based on fine grid technology. *Sci. Data* **2025**, *12* (1), 640.

(3) Yuan, J.-H.; Lyon, T. P. Promoting global CCS RDD&D by stronger U.S.−China collaboration. *Renew. Sustain. Energy Rev.* **2012**, *16* (9), 6746−6769.

(4) Liu, Z.; Guan, D.; Wei, W.; Davis, S. J.; Ciais, P.; Bai, J.; Peng, S.; Zhang, Q.; Hubacek, K.; Marland, G.; Andres, R. J.; Crawford-Brown, D.; Lin, J.; Zhao, H.; Hong, C.; Boden, T. A.; Feng, K.; Peters, G. P.; Xi, F.; Liu, J.; Li, Y.; Zhao, Y.; Zeng, N.; He, K. Reduced carbon emission estimates from fossil fuel combustion and cement production in China. *Nature* **2015**, *524* (7565), 335−338.

(5) Mallapaty, S. How China could be carbon neutral by mid-century. *Nature* **2020**, *586* (7830), 482−483.

(6) Moore, E. A.; Chua, A.; Middleton, E.; Middleton, R.; Sale, K.; Mehdi, Q.; Miranda, M.; Grol, E.; Azarijafari, H.; Kirchain, R. How can the Cement Industry Enable Industrial Decarbonization at Scale? *Environ. Sci. Technol.* **2025**, *59* (36), 19233−19243.

(7) Bachu, S.; Bonijoly, D.; Bradshaw, J.; Burruss, R.; Holloway, S.; Christensen, N. P.; Mathiassen, O. M. CO_2 storage capacity estimation: Methodology and gaps. *Int. J. Greenhouse Gas Control* **2007**, *1* (4), 430−443.

(8) Bradshaw, J.; Bachu, S.; Bonijoly, D.; Burruss, R.; Holloway, S.; Christensen, N. P.; Mathiassen, O. M. CO_2 storage capacity estimation: issues and development of standards. *Int. J. Greenhouse Gas Control* **2007**, *1*, 62−68.

(9) Anderson, S. T. Cost Implications of Uncertainty in CO_2 Storage Resource Estimates: A Review. *Nat. Resour Res.* **2017**, *26* (2), 137−159.

(10) Elenius, M.; Skurtveit, E.; Yarushina, V.; Baig, I.; Sundal, A.; Wangen, M.; Landschulze, K.; Kaufmann, R.; Choi, J. C.; Hellevang, H.; Podladchikov, Y.; Aavatsmark, I.; Gasda, S. E. Assessment of CO_2 storage capacity based on sparse data: Skade Formation. *Int. J. Greenhouse Gas Control* **2018**, *79*, 252−271.

(11) Wei, N.; Jiao, Z.; Ellett, K.; Ku, A. Y.; Liu, S.; Middleton, R.; Li, X. Decarbonizing the Coal-Fired Power Sector in China via Carbon Capture, Geological Utilization, and Storage Technology. *Environ. Sci. Technol.* **2021**, *55* (19), 13164−13173.

(12) Wei, N.; Liu, S. N.; Jiao, Z. S.; Li, X. C. A possible contribution of carbon capture, geological utilization, and storage in the Chinese crude steel industry for carbon neutrality. *J. Cleaner Prod.* **2022**, *374*, 133793.

(13) Li, X.; Wei, N.; Liu, Y.; Fang, Z.; Dahowski, R. T.; Davidson, C. L. CO_2 point emission and geological storage capacity in China. *Energy Procedia* **2009**, *1* (1), 2793−2800.

(14) Wei, N.; Li, X.; Fang, Z.; Bai, B.; Li, Q.; Liu, S.; Jia, Y. Regional resource distribution of onshore carbon geological utilization in China. *J. CO2 Util.* **2015**, *11*, 20−30.

(15) Liu, S.; Wei, N.; Jiang, D.; Nie, L.; Cai, B.; Tao, Y.; Li, X. Emission reduction path for coal-based enterprises via carbon capture, geological utilization, and storage: China energy group. *Energy* **2023**, *273*, 127222.

(16) Wang, S.; Vincent, C. J.; Stephenson, M. H.; Zeng, R. Assessment of storage capacity for CO_2 in saline aquifers near hydrocarbon fields, northern Songliao Basin, China. *Greenhouse Gases:Sci. Technol.* **2014**, *4* (3), 366−383.

(17) Sun, L.; Dou, H.; Li, Z.; Hu, Y.; Hao, X. Assessment of CO_2 storage potential and carbon capture, utilization and storage prospect in China. *J. Energy Inst.* **2018**, *91* (6), 970−977.

(18) Heinemann, N.; Haszeldine, R. S.; Shu, Y.; Stewart, R. J.; Scott, V.; Wilkinson, M. CO2 sequestration with limited sealing capability: A new injection and storage strategy in the Pearl River Mouth Basin (China). *Int. J. Greenhouse Gas Control* **2018**, *68*, 230−235.

(19) Zhou, D.; Li, P.; Liang, X.; Liu, M.; Wang, L. A long-term strategic plan of offshore CO_2 transport and storage in northern South China Sea for a low-carbon development in Guangdong province, China. *Int. J. Greenhouse Gas Control* **2018**, *70*, 76−87.

(20) Zhou, D.; Zhao, Z.; Liao, J.; Sun, Z. A preliminary assessment on CO_2 storage capacity in the Pearl River Mouth Basin offshore Guangdong, China. *Int. J. Greenhouse Gas Control* **2011**, *5* (2), 308−317.

(21) Qiao, X.; Li, G.; Li, M.; Wang, Z. CO_2 storage capacity assessment of deep saline aquifers in the Subei Basin, East China. *Int. J. Greenhouse Gas Control* **2012**, *11* (0), 52−63.

(22) Zhang, X.; Li, K.; Wei, N.; Li, Z.; Fan, J.-L. Advances, challenges, and perspectives for CCUS source-sink matching models under carbon neutrality target. *J. Carbon Neutr.* **2022**, *1* (1), 12.

(23) Lin, K.; Wei, N.; Jiang, D.; Zhang, Y. Progress and perspectives in source-sink matching for CCUS: A critical review. *Renew. Sustain. Energy Rev.* **2026**, *226*, 116313.

(24) Middleton, R. S.; Yaw, S. The cost of getting CCS wrong: Uncertainty, infrastructure design, and stranded CO_2. *Int. J. Greenhouse Gas Control* **2018**, *70*, 1−11.

(25) DOE-NETL. *Site Screening, Selection and Characterization for Storage of CO2 in Deep Geologic Formations*; United State Department of Energy: National Energy Technology Laboratory, https://www.netl.doe.gov/sites/default/files/2018-10/BPM-SiteScreening.pdf: Albany, Oregon, 2017.

(26) Zhai, H.; Ou, Y.; Rubin, E. S. Opportunities for Decarbonizing Existing U.S. Coal-Fired Power Plants via CO_2 Capture, Utilization and Storage. *Environ. Sci. Technol.* **2015**, *49* (13), 7571−7579.

(27) IEA *CO_2 Storage Resources and Their Development*; 2022.

(28) Bachu, S. Review of CO_2 storage efficiency in deep saline aquifers. *Int. J. Greenhouse Gas Control* **2015**, *40*, 188−202.

(29) Huo, C. *Study on the Potential Evaluation and the Storage Areas of the Carbon Dioxide Seabed Storage in Offshore China*; Dalian Maritime University: Dalian, 2014.

(30) Sun, L. L.; Liu, Q.; Chen, H. J.; Yu, H.; Li, L.; Li, L. T.; Li, Y. Z.; Adenutsi, C. D. Source-sink matching and cost analysis of offshore carbon capture, utilization, and storage in China. *Energy* **2024**, *291*, 130137.

(31) Wei, N.; Li, X. C.; Wang, Y.; Dahowski, R. T.; Davidson, C. L.; Bromhal, G. S. A preliminary sub-basin scale evaluation framework of site suitability for onshore aquifer-based CO_2 storage in China. *Int. J. Greenhouse Gas Control* **2013**, *12*, 231−246.

(32) Li, X.; Wei, N.; Jiao, Z.; Liu, S.; Dahowski, R. Cost curve of large-scale deployment of CO_2-enhanced water recovery technology in modern coal chemical industries in China. *Int. J. Greenhouse Gas Control* **2019**, *81*, 66−82.

(33) Birkholzer, J. T.; Oldenburg, C. M.; Zhou, Q. CO_2 migration and pressure evolution in deep saline aquifers. *Int. J. Greenhouse Gas Control* **2015**, *40*, 203−220.

(34) Chadwick, R. A.; Williams, G. A.; Falcon-Suarez, I. Forensic mapping of seismic velocity heterogeneity in a CO_2 layer at the Sleipner CO_2 storage operation, North Sea, using time-lapse seismics. *Int. J. Greenhouse Gas Control* **2019**, *90*, 102793.

(35) Deng, H. L.; Stauffer, P. H.; Dai, Z. X.; Jiao, Z. S.; Surdam, R. C. Simulation of industrial-scale CO_2 storage: Multi-scale heterogeneity and its impacts on storage capacity, injectivity and leakage. *Int. J. Greenhouse Gas Control* **2012**, *10*, 397−418.

(36) Alcalde, J.; Heinemann, N.; James, A.; Bond, C. E.; Ghanbari, S.; Mackay, E. J.; Haszeldine, R. S.; Faulkner, D. R.; Worden, R. H.; Allen, M. J. A criteria-driven approach to the CO_2 storage site selection of East Mey for the acorn project in the North Sea. *Mar. Pet. Geol.* **2021**, *133*, 105309.

(37) Ye, J.; Afifi, A.; Rowaihy, F.; Baby, G.; De Santiago, A.; Tasianas, A.; Hamieh, A.; Khodayeva, A.; Al-Juaied, M.; Meckel, T. A.; Hoteit, H. Evaluation of geological CO_2 storage potential in Saudi Arabian sedimentary basins. *Earth-Sci. Rev.* **2023**, *244*, 104539.

(38) Cantucci, B.; Buttinelli, M.; Procesi, M.; Sciarra, A.; Anselm, M. Algorithms for CO_2 Storage Capacity Estimation: Review and Case Study.In *Geologic Carbon Sequestration*; Springer: Cham, 2016; pp 21−44.

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184

(39) Bachu, S. *Comparison between Methodologies Recommended for Estimation of CO_2 Storage Capacity in Geological Media by the CSLF Task Force on CO_2 Storage Capacity Estimation and the USDOE Capacity and Fairways Subgroup of the Regional Carbon Sequestration Partnerships Program*, 2008.
(40) Pruess, K.; García, J.; Kovscek, T.; Oldenburg, C.; Rutqvist, J.; Steefel, C.; Xu, T. Code intercomparison builds confidence in numerical simulation models for geologic disposal of CO_2. *Energy* **2004**, *29* (9−10), 1431−1444.
(41) Goodman, A.; Bromhal, G.; Strazisar, B.; Rodosta, T.; Guthrie, W. F.; Allen, D.; Guthrie, G. Comparison of methods for geologic storage of carbon dioxide in saline formations. *Int. J. Greenhouse Gas Control* **2013**, *18*, 329−342.
(42) Popova, O. H.; Small, M. J.; McCoy, S. T.; Thomas, A. C.; Rose, S.; Karimi, B.; Carter, K.; Goodman, A. Spatial Stochastic Modeling of Sedimentary Formations to Assess CO_2 Storage Potential. *Environ. Sci. Technol.* **2014**, *48* (11), 6247−6255.
(43) De Silva, P. N. K.; Ranjith, P. G. A study of methodologies for CO_2 storage capacity estimation of saline aquifers. *Fuel* **2012**, *93*, 13−27.
(44) Celia, M. A.; Bachu, S.; Nordbotten, J. M.; Bandilla, K. W. Status of CO_2 storage in deep saline aquifers with emphasis on modeling approaches and practical simulations. *Water Resour. Res.* **2015**, *51* (9), 6846−6892.
(45) Rose, K. *Development of an Open Global Oil and Gas Infrastructure Inventory and Geodatabase*: United States, 2018.
(46) Khosravi, M.; Askari, Z.; Okwen, R. T.; Taft, K. Evaluating the regional geological characteristics of the St. Peter Sandstone and Everton Formation for CO_2 storage in Southern Illinois: A case study on site-specific injection feasibility in Washington County, Illinois. *Int. J. Greenhouse Gas Control* **2025**, *141*, 104292.
(47) Zhang, Y.-F.; Qu, M.-L.; Yang, J.-P.; Foroughi, S.; Niu, B.; Yu, Z.-T.; Gao, X.; Blunt, M. J.; Lin, Q. Prediction of CO_2 storage efficiency and its uncertainty using deep-convolutional GANs and pore network modelling. *Appl. Energy* **2025**, *381*, 125142.
(48) Edouard, M. N.; Okere, C. J.; Ejike, C.; Dong, P. C.; Suliman, M. A. M. Comparative numerical study on the co-optimization of CO_2 storage and utilization in EOR, EGR, and EWR: Implications for CCUS project development. *Appl. Energy* **2023**, *347*, 121448.
(49) Mathias, S. A.; Gluyas, J. G.; Goldthorpe, W. H.; Mackay, E. J. Impact of Maximum Allowable Cost on CO_2 Storage Capacity in Saline Formations. *Environ. Sci. Technol.* **2015**, *49* (22), 13510−13518.
(50) Knoope, M. M. J.; Guijt, W.; Ramírez, A.; Faaij, A. P. C. Improved cost models for optimizing CO_2 pipeline configuration for point-to-point pipelines and simple networks. *Int. J. Greenhouse Gas Control* **2014**, *22*, 25−46.
(51) McCoy, S. T.; Rubin, E. S. An engineering-economic model of pipeline transport of CO_2 with application to carbon capture and storage. *Int. J. Greenhouse Gas Control* **2008**, *2* (2), 219−229.
(52) Middleton, R. S.; Bielicki, J. M. A scalable infrastructure model for carbon capture and storage. *Energy Policy* **2009**, *37* (3), 1052−1060.
(53) Leeson, D.; Mac Dowell, N.; Shah, N.; Petit, C.; Fennell, P. S. A Techno-economic analysis and systematic review of carbon capture and storage (CCS) applied to the iron and steel, cement, oil refining and pulp and paper industries, as well as other high purity sources. *Int. J. Greenhouse Gas Control* **2017**, *61*, 71−84.
(54) Zimmermann, A. W.; Wunderlich, J.; Müller, L.; Buchner, G. A.; Marxen, A.; Michailos, S.; Armstrong, K.; Naims, H.; McCord, S.; Styring, P.; Sick, V.; Schomäcker, R. Techno-Economic Assessment Guidelines for CO_2 Utilization. *Front. Energy Res.* **2020**, *8*, 5.
(55) Bui, M.; Adjiman, C. S.; Bardow, A.; Anthony, E. J.; Boston, A.; Brown, S.; Fennell, P. S.; Fuss, S.; Galindo, A.; Hackett, L. A.; Hallett, J. P.; Herzog, H. J.; Jackson, G.; Kemper, J.; Krevor, S.; Maitland, G. C.; Matuszewski, M.; Metcalfe, I. S.; Petit, C.; Puxty, G.; Reimer, J.; Reiner, D. M.; Rubin, E. S.; Scott, S. A.; Shah, N.; Smit, B.; Trusler, J. P. M.; Webley, P.; Wilcox, J.; Mac Dowell, N. Carbon capture and storage (CCS): the way forward. *Energy Environ. Sci.* **2018**, *11* (5), 1062−1176.
(56) Leung, D. Y. C.; Caramanna, G.; Maroto-Valer, M. M. An overview of current status of carbon dioxide capture and storage technologies. *Renewable Sustainable Energy Rev.* **2014**, *39*, 426−443.
(57) Rowaihy, F.; Hamieh, A.; Odeh, N.; Hejazi, M.; Al-Juaied, M.; Afifi, A. M.; Hoteit, H. Decarbonizing Saudi Arabia energy and industrial sectors: Assessment of carbon capture cost. *Carbon Capture Sci. Technol.* **2025**, *14*, 100375.
(58) Wei, N.; Lin, K. Y.; Liu, S. N.; Wang, W. D.; Cheng, G. Q.; Yang, L. X.; Jiao, Z. S.; Li, X. C. A Microflow and Bioinspired Dendritic Topology Optimization for CO_2 Capture, Utilization and Storage Hubs in China: Integrating Microfluid Dynamics with Macro-Scale Infrastructure Design. *ACS Sustainable Chem. Eng.* **2026**, *14* (9), 4403−4421.
(59) Wei, N.; Li, X.; Jiao, Z.; Stauffer, P. H.; Liu, S.; Ellett, K.; Middleton, R. S. A Hierarchical Framework for CO_2 Storage Capacity in Deep Saline Aquifer Formations. *Front. Earth Sci.* **2022**, *9*, 777323.
(60) Rubin, E. S.; Davison, J. E.; Herzog, H. J. The cost of CO_2 capture and storage. *Int. J. Greenhouse Gas Control* **2015**, *40*, 378−400.
(61) *EIA Oil and Gas Lease Equipment and Operating Costs 1994 Through 2009*; Energy Information Administration, Department of Energy: U.S, 2010.
(62) Mi, L. *Current status of global CO_2 ocean sequestration and opportunities and challenges in China offshore areas*. CHINA OFFSHORE OIL AND GAS (GEOLOGY) 2023, vol *35*, pp 123−135.
(63) Hamieh, A.; Rowaihy, F.; Al-Juaied, M.; Abo-Khatwa, A. N.; Afifi, A. M.; Hoteit, H. Quantification and analysis of CO_2 footprint from industrial facilities in Saudi Arabia. *Energy Convers. Manage.:X* **2022**, *16*, 100299.

Downloaded from pubs.acs.org/esthag/article-pdf/60/27/19170/6529877/1/acs.est.5c14041.pdf by guest on 25 October 2026

https://doi.org/10.1021/acs.est.5c14041
Environ. Sci. Technol. 2026, 60, 19170−19184